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Screening Pilot

CBAS-R Fall Screening Pilot: Technical Report on Psychometric and Practical Feasibility

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Abstract

The Computer-Based Assessment System in Early Reading (CBAS-R) was implemented with 287 students in kindergarten through second grade at two schools in early fall. The three parameter Item Response Theory (3P IRT) computer adaptive test administered up to forty items using maximum likelihood estimation to pilot the CBAS-R as screening tool. Results are discussed in terms of psychometric and practical feasibility for this purpose. Specific research questions addressed precision, efficiency, content validity, criterion validity, response function, and timing. Findings support feasibility of the CBAS-R for screening, and psychometric soundness is largely maintained when fewer items are administered.

Keywords: screening pilot, fall 2008, CBAS-R

Research Log

Activity	Lead Person	Timeframe
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CBAS-R Fall Screening Pilot:

Technical Report on Psychometric and Practical Feasibility

The Computer-Based Assessment System in Early Reading (CBAS-R) project aims to meet an important need for efficient, precise, and instructionally linked assessment of early reading level and rate of growth. The CBAS-R is a three parameter item response theory (3-P IRT) computer adaptive test (CAT). The study described here operated as a pilot study of a fall screening application of the developing CBAS-R measure.

Need

In elementary education, use of neither Computer Adaptive Testing (CAT) nor Item Response Theory (IRT) is substantially established. The benefits of CAT that are most often cited in the professional literature include: (a) individualized dynamic assessment, which does not rely on a fixed set of items across administrations or individuals; (b) testing time that is reduced by one-half to one-third (or more) of traditional tests because irrelevant items are excluded from the administration; (c) test applicability and measurement precision across a broad range of skills/abilities, and (d) more precise methods to equate assessment outcomes across alternate forms or administrations (Kingsbury & Houser, 1999; Weiss, 2004a; Zickar, Overton, Taylor, & Harms, 1999).

Students at-risk for reading deficits and learning disabilities can be identified as early as kindergarten (Felton & Pepper, 1995; Lennon & Slesinski, 1999; Pepper & Felton, 1995) and first grade (O'Connor & Jenkins, 1999). Identification should lead to intervention, and such intervention should be hinged to formative assessment. The use of formative assessment data to guide instruction contributes to more effective instruction and higher student achievement (Jinkins, 2001; Taylor, Pearson, Clark, & Walpole, 2000a, 2000b; Taylor, Pearson, Peterson, &

Rodriguez, 2003, 2005). Effective use of formative assessment can improve student achievement by .5 standard deviations (Black & Wiliam, 1998).

Teachers need efficient and instructionally useful measures of early reading achievement for students within their classrooms. Particularly within a Response to Intervention (RtI) framework, measures that are efficient and precise for both screening and progress monitoring are particularly important.

Purpose

The goals of the Computer-Based Assessment System in early Reading (CBAS-R) are to offer efficient classroom-based computer adaptive assessment for purposes of screening, skills diagnosis, and progress monitoring of early reading. The purpose of the study described herein is to examine a pilot fall screening administration of the CBAS-R for psychometric and practical feasibility. Aspects of feasibility considered are specified here as research questions.

Research Questions

1. Precision and efficiency: How precise is the CBAS-R ability estimate, and how does this differ across the range of ability estimates? How few items need to be administered in order to achieve a minimally acceptable level of precision ($SE \leq .20$)?
2. Content validity: How well does the CBAS-R test represent the content of the dynamic construct of reading proficiency across stages of development? How does content of the test adjust across the range of ability estimates?
3. Criterion validity: How valid is the CBAS-R ability estimate? Does it appropriately discriminate students across grade levels? Does it correlate to criterion measures, including teacher ratings and reading inventories?

4. Response function: To what extent is the response environment of the test (percent correct) equalized across the range of ability estimates?
5. Timing: How much time do student examinees take on the assessment?

Method

Participants

Students in kindergarten through grade 2 were drawn from two Minnesota schools, one suburban and one urban. School demographics are presented in Table 1. Sample sizes by school and grade, including special education and English Language Learner (ELL) status, are presented in Table 2. Overall, 287 participants took the CBAS-R for fall screening, with 8% of these receiving special education and 12% receiving ELL services.

Table 1: *School Demographics*

Category	School A	School B
White	94%	12%
Black	<1%	61%
Hispanic	<1%	12%
Asian/Pacific Islander	5%	14%
American Indian/Alaskian Native	<1%	2%
Free and Reduced Lunch	13%	83%

Table 2: *Participants by School, Grade, and Special Education or ELL status*

School	Grade	Number of Students	SPED	ELL
School A	K	34	1	0
	1	55	3	0
	2	75	0	0
School A Total		164	4	0
School B	K	42	5	17
	1	40	5	6
	2	41	9	11
School B Total		123	19	34
GRAND TOTAL		287	23 (8%)	34 (12%)

Measures and Procedures

CBAS-R

The CBAS-R is an adaptive test which selects a next item from a large pool of candidate items (n = 388). Each item in the pool exists in exactly one of six skill domains which align to the framework proposed by the National Reading Panel. In Table 3, the items in each domain are described across three parameters: a = discrimination, b=difficulty, c=guessing parameter. Because the items are selected based on maximum information at the current theta estimate, Table 3 also describes the distribution of thetas at which items in each domain offer maximum information.

Table 3: *Descriptive Statistics for CBAS Screening CAT Item Pool*

		a	b			c	Theta of max info
Domain	n	**M** (SD)	**M** (SD)	Min	Max	**M** (SD)	**M** (SD)
Concepts of Print	37	**1.91** (0.23)	**-1.30** (0.60)	-2.58	0.54	**0.17** (0.03)	**-1.22** (0.60)
Phonemic Awareness	50	**1.72** (0.18)	**-0.70** (0.79)	-1.90	1.02	**0.20** (0.05)	**-0.61** (0.79)
Phonics	96	**1.91** (0.21)	**-0.06** (1.05)	-1.90	2.47	**0.17** (0.04)	**0.01** (1.05)
Fluency & Decoding	97	**2.00** (0.27)	**0.07** (0.82)	-1.13	1.99	**0.17** (0.04)	**0.15** (0.82)
Vocabulary	42	**1.86** (0.21)	**1.08** (0.89)	-2.04	3.00	**0.17** (0.04)	**1.16** (0.89)
Comprehension	66	**1.85** (0.23)	**1.05** (0.59)	0.16	2.91	**0.19** (0.05)	**1.13** (0.59)

Participants took the CBAS tests on laptop computers set up within their school, using a mouse to respond to the visual and auditory stimuli. The participants wore padded headphones and faced the computer screen, which was surrounded by a privacy carrel. The keyboard was covered and not used during the assessment. Two CBAS-R project staff (certified teachers) proctored the administration of 12 students at a time.

Test module configuration: During the test, item numbers were displayed (e.g. 1 through 40); a 20 minute time limit was set. No omissions were allowed in answering. For each presented item, the participant had to select an answer in order to enable the arrow signifying "next item;" after clicking on the arrow, participants could not return to previously answered items.

Item selection: The initial theta estimate was set at zero. For the initial item and for each next item administered, the program selected the item with maximal item information at the current theta estimate. Individuals' current theta estimate was determined by maximum likelihood estimation. After the first item, then, each item presented was selected adaptively based on previous responses within the occurring module.

Criterion measures

Validity criteria include a project- administered measure (the Teacher Rating Scale) and existing school reading data. Each school used different reading measures, to which we correlate the theta estimates from the CBAS-R experimental measures above. Criterion measures include the following:

Teacher Rating Scale for Students' Reading Proficiency. In most participating classrooms, the teacher completed a rating scale. All students in the class were listed, and the teacher was asked to rate their individual overall reading proficiency on a seven point scale, relative to one another. The Teacher Rating Scale is included in Appendix A.

Fountas & Pinnell Benchmark Assessment System. On this assessment, students read aloud a passage at their estimated level and word accuracy is recorded. Comprehension questions are asked after the reading. Students are assigned an instructional level based on

performance within a range of accuracy and comprehension. Instructional levels B through M correspond to the range of typical grade 1 and 2 reading proficiency.

Phonological Awareness Literacy Screening (PALS): PALS is a group- and individually-administered measure of students' literacy knowledge in the domains of phonological awareness, alphabet knowledge, knowledge of letter sounds, spelling, concept of word, word recognition in isolation, and oral passage reading. The criterion score used in this study is a summed score derived from performance in all domains.

Results

Question 1. Precision and efficiency

How precise is the CBAS-R ability estimate, and how does this differ across the range of ability estimates? How few items need to be administered in order to achieve a minimally acceptable level of precision (SE $\leq$ .20)?

Before reviewing findings pertinent to precision and efficiency, we clarify: *What is precision within an IRT CAT?* Reliability usually refers to the precision of measurement or the degree to which measurement is free of error. In classical test theory this is measured using a single index, the ratio of true and observed score variance. This index makes it possible to determine the test's average reliability, for example in order to compare two tests. In IRT, precision is not uniform across the entire range of test scores; instead, item and test information are incorporated. Information is a function of the model parameters. In a 3PL IRT model, the information function incorporates the three item parameters, where *a* is the discrimination parameter, *b* is the difficulty parameter, c is the probability that the student with the lowest theta will get the item correct (also called the guessing parameter). In the equation for information that follows, $P(\theta)$ is the probability that a student with a given theta estimate will get the item

correct, and *Q(θ)* is the probability that a student with a given theta will get the item incorrect:

$$I_i(\theta) = a^2\left[\frac{Q_i(\theta)}{P_i(\theta)}\right]\left[\frac{Pi(\theta) - c^2}{\left(1 - c^2\right)}\right]$$

The standard error of measurement, which is the estimated standard deviation of error, is the reciprocal of the square root of the test information given at the given ability level:

$SE(\theta) = \frac{1}{\sqrt{I(\theta)}}$. Therefore, $I(\theta) = \frac{1}{SE^2}$. Consequently, more information implies less error of measurement. When measurement error is reduced, reliability of the test increases. Thus, standard error values within IRT give an estimate of the reliability of an assessment.

In examining efficiency of precision, one of the goals is to identify the fewest amount of items that can be administered while maintaining standard error values that are below the pre-determined cutoff value of 0.2 .Figures 1-4, depicted below, represent the standard error for administrations of 40, 30, 20 and 15 items. The CBAS-R was administered with termination criteria set at 40 items in live data collection; the terminations at fewer items represent post hoc, artificial terminations in which only responses from the first number of items given are considered from each student's actual performance.

For all figures, standard error values are not shown beyond approximately 1.3 because students who took this test did not receive theta values greater than 1.3. In Figure 1, the standard error is shown when 40 items were administered. For most theta values between -2 and 1.2, standard error remains below the desired value of 0.2. The only peak above 0.2 occurs at a theta value of approximately 1.5.

Figure 2 shows the standard error of measurement for an administration of 30 items. Although standard error values are approaching 0.2, values remain below 0.2 for most theta values between -2 and 1.4.

Figure 3 displays the standard error when termination was set at 20 items. There are spikes at -1.8, -0.5 and 0.6 where the standard error values exceed 0.2. However, standard error is generally at or below 0.2 for most theta values ranging from approximately -1.6 through 1.3.

Figure 4 shows the standard error when 15 items were administered to students. For theta values between -2.5 and 2, standard error was generally in the range of 0.2 to 0.5. Standard error is highest for theta values starting from -2.8 and approaching -3. Because reducing the number of items administered to just 15 items leaves standard error above .2, we hereafter only consider the case where CBAS-R terminates at 20 items as a minimally precise test with maximal time efficiency.

Figures 1-4 revealed that standard error is highest (between 0.6 and 1) for theta values less than -2.7. Therefore, when 40, 30, 20 and 15 items are administered, our screening test is not as precise of a measure for theta values less than -2.7. These figures also reveal that administering 20 items does not significantly increase standard error when compared to 40 items. For 20 items, standard error maintains values below 0.2. However, when 15 items were administered, the standard error exceeded 0.2 for most theta values. These results imply that subsequent administrations of the CBAS-R can terminate after as few as 20 items while maintaining standard error values below 0.2.

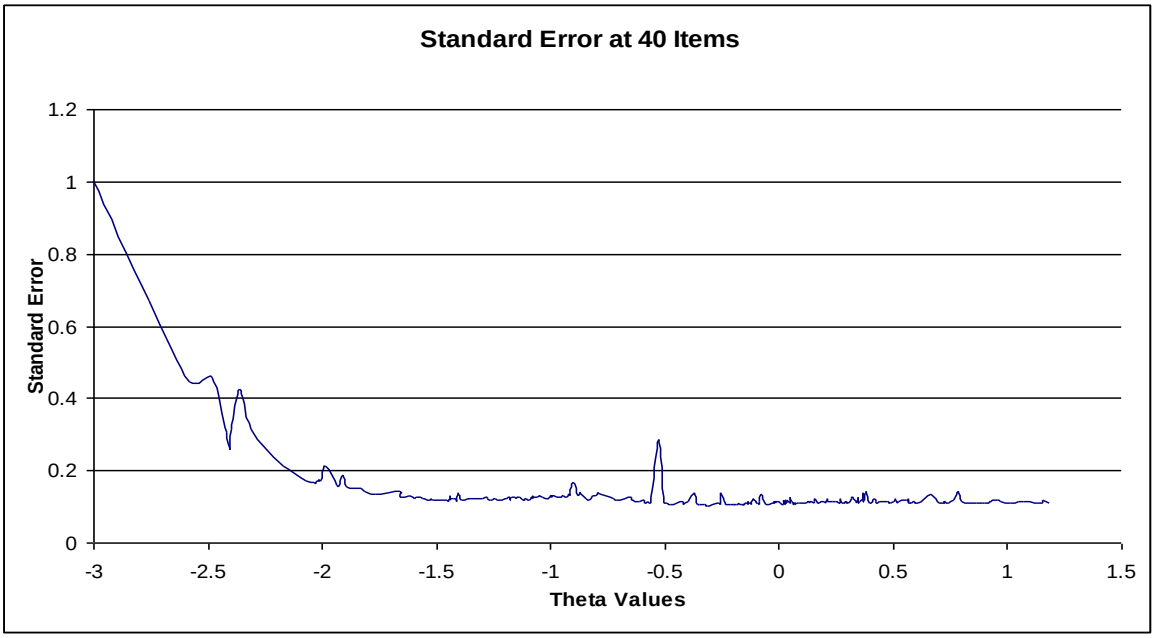


Figure 1. Standard Error Values for a 40 Item Test Administration.

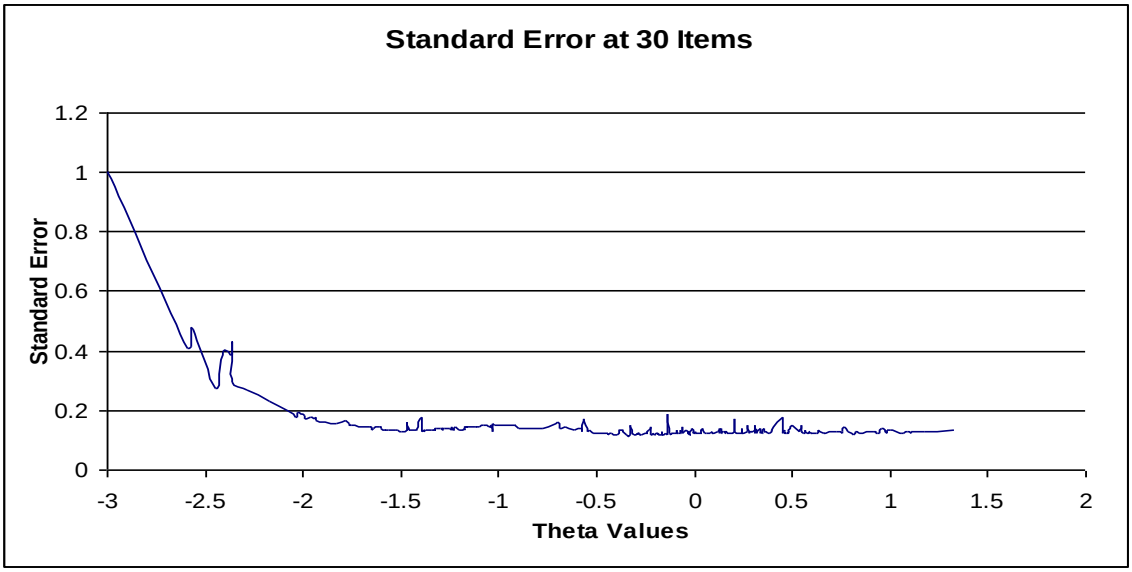


Figure 2. Standard Error Values for a 30 Item Test Administration.

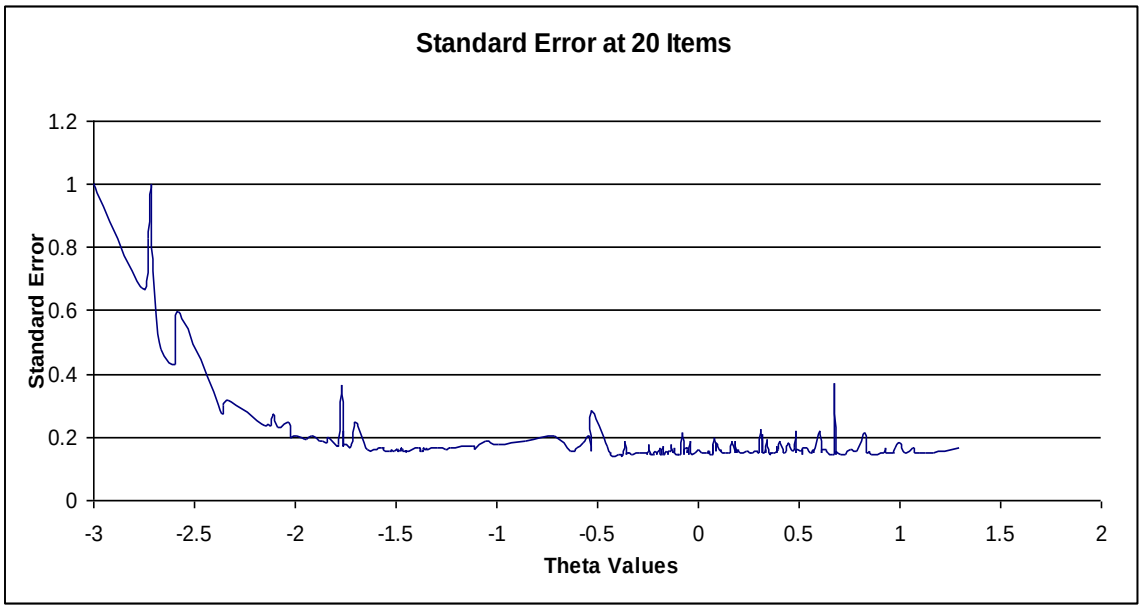


Figure 3. Standard Error Values for a 20 Item Test Administration.

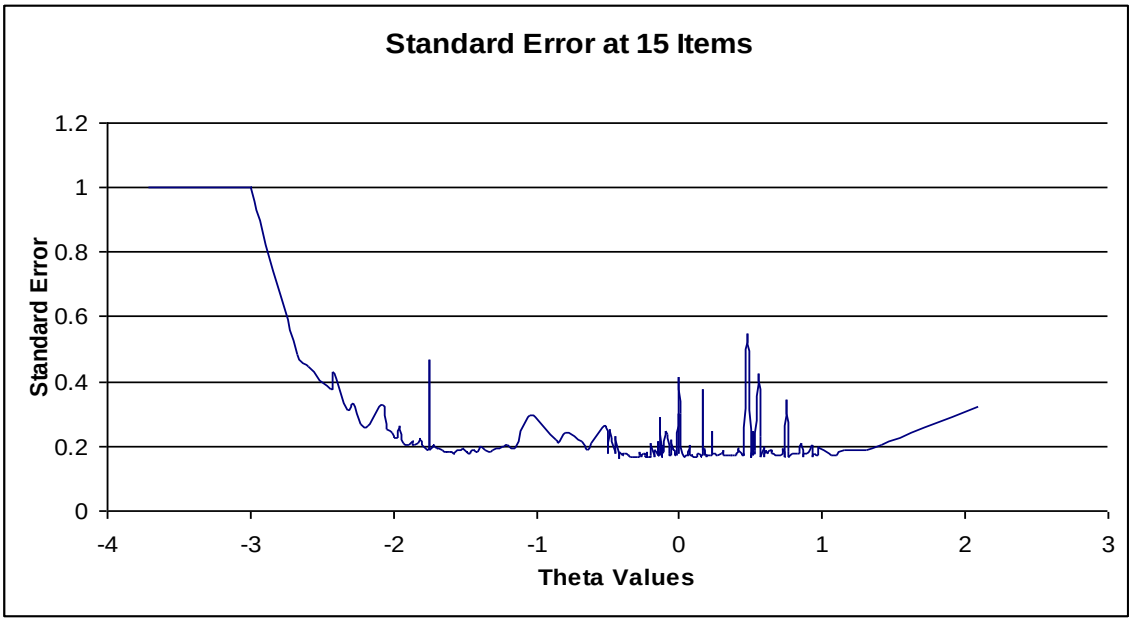


Figure 4. Standard Error Values for a 15 Item Test Administration.

Information functions

Figures 5-6 display test information functions for the CBAS-R. Since 20 items were acceptable for standard error purposes, the test information functions (and all further results for additional research questions below) will be compared for administrations of 40 and 20 items.

In Figure 5, the test information function is shown for an administration with termination at 40 items. Overall, the information function is highest for students with ability between -0.4 and -0.1. Generally, the function is higher for theta values greater than -0.3.

Figure 6 shows the test information function when termination of the test is set at 20 items (post hoc). The information function appears to be highest for students with an ability of about -0.4 to -0.3. As with 40 items, the function is higher for theta values greater than -0.3.

As expected, Figures 5-6 correspond with Figures 1 and 4 in that information decreases as standard error increases, which confirms the inverse relationship between the two variables. Thus, for both 40 and 20 items, barely any information is obtained for theta values less than -2.5. Clearly, administering 40 items yields an information function with a higher range of values, the maximum being 90; whereas, with 20 items, the maximum information given is 50. Administering half as many items decreased information by a factor of about 2 (specifically, 1.8); however the trend of the function is similar (gradually increases for higher theta values).

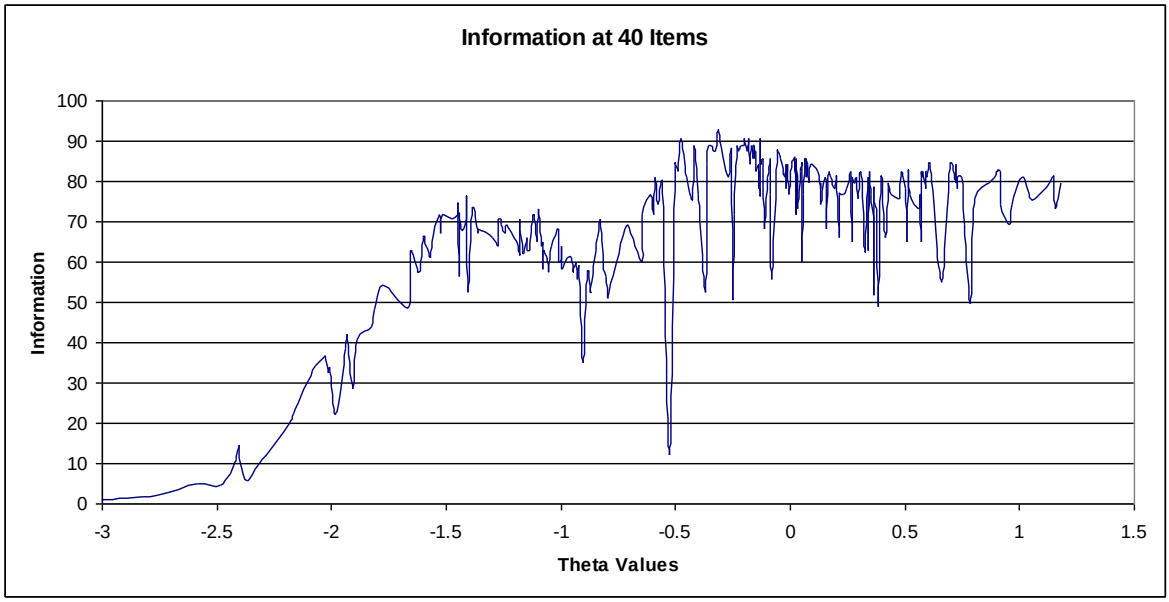


Figure 5. Test Information Function for a Test Administration of 40 Items.

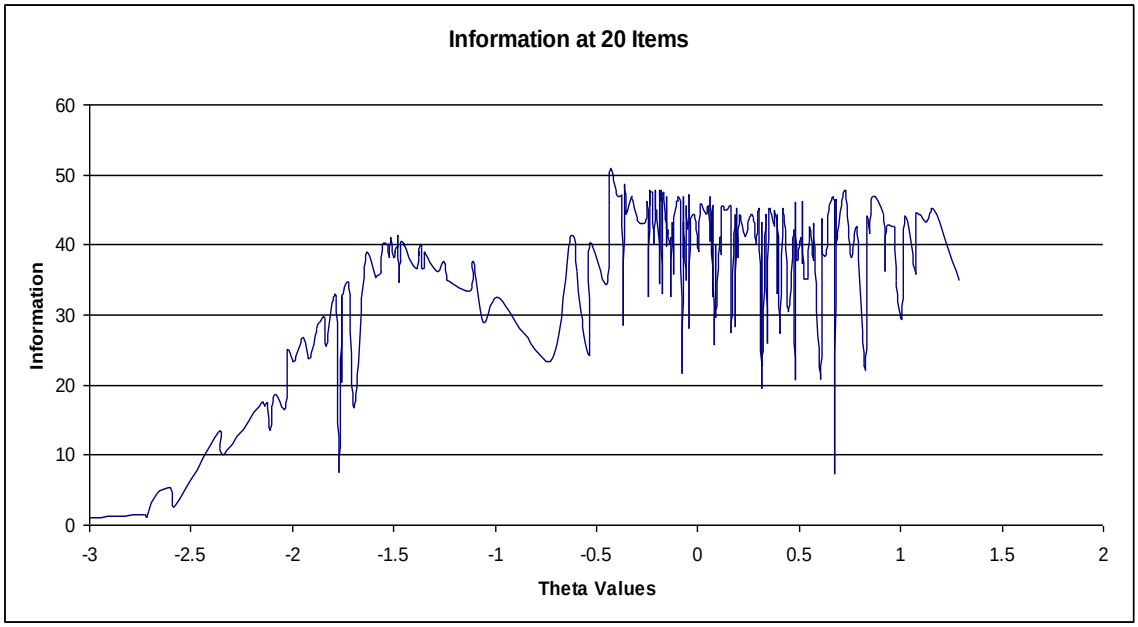


Figure 6. Test Information Function for a Test Administration of 20 Items.

Question 2. Content validity

How well does the CBAS-R test represent the content of the dynamic construct of reading proficiency across stages of development? How does content of the test adjust across the range of ability estimates?

Content validity for an individually adaptive reading measure must be demonstrated dynamically across the primary grades and across levels of proficiency observed within those years. Content in literacy instruction for emergent readers, for example, differs substantively from the content of instruction for fluent readers. While the CBAS-R measures a unified construct (reading) across these years, the five skill domains carry different weight at each stage of early reading development.

Figures 7 and 8 show patterns in item content presented on the adaptive CBAS-R across the spectrum of reading proficiency. These data are presented in Tables 4 and 5 as well, categorized into bins by final CBAS-R abilitye (theta) estimate.

In Figure 7, the CBAS-R final theta estimate (horizontal axis) is based on termination at 40 items. Items within each skill domain have a unimodal frequency distribution across the range of final thetas achieved by the sample. Among the five domains, Phonemic Awareness has its peak at the lowest final theta estimate; Comprehension peaks at the highest final theta. The distribution for Comprehension fails to approach normal because the upper bound of participants' final theta in this sample is lower than the theta of maximum information for many comprehension items (see Table 3 in Methods).

Figure 8 demonstrates the same patterns in item content presented, as they differ across the spectrum of final overall performance, for termination at 20 items. Frequency distributions

for each of the domains are of similar shape at 20 items and 40 items, with the horizontal scale differing accordingly.

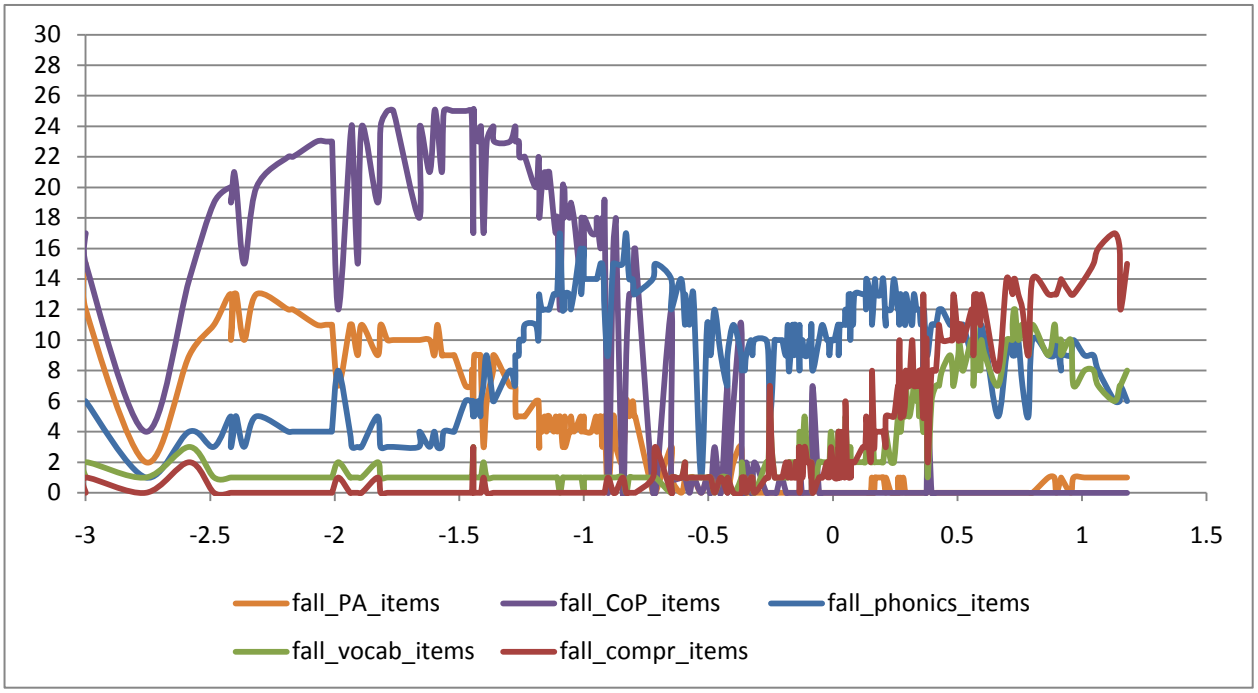


Figure 7. Number of Items Administered in Each Domain, by Final Theta (termination = 40 items)

Table 4. *Mean Number of Items from Each Domain, by Final Theta (termination = 40 items)*

Final theta	n	Concepts of Print	Phonemic Awareness	Phonics	Vocab.	Comp.
-3.0 to -2.0	19	18.95	11.32	4.16	1.16	0.16
-1.99 to -1.0	65	21.16	6.69	8.23	1.06	0.09
-0.99 to 0	93	4.91	1.33	11.07	1.41	0.97
.01 to 1.0	103	0.06	.015	10.96	5.91	7.56
1.01 to 2.0	7	0	1.00	7.29	7.29	15.00

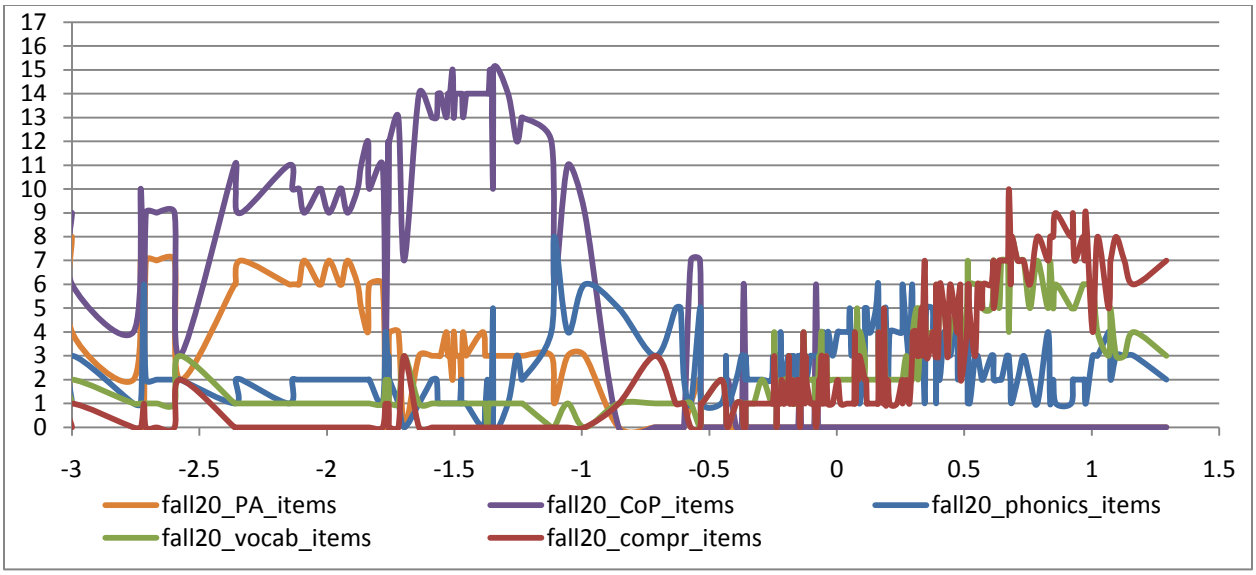


Figure 8. Number of Items Administered in Each Domain, by Final Theta (termination = 20 items)

Table 5. *Mean Number of Items from Each Domain, by Final Theta (termination = 20 items)*

Final theta	n	Concepts of Print	Phonemic Awareness	Phonics	Vocabulary	Comprehension
-3.0 to -2.0	21	8.57	5.71	1.95	1.14	0.19
-1.99 to -1.0	87	12.95	3.28	1.45	1.01	0.06
-0.99 to 0	75	0.57	0.11	2.65	1.53	1.24
.01 to 1.0	95	0	0	3.27	4.14	4.19
1.01 to 2.0	9	0	0	3.00	3.89	6.56

Content distribution varies across grade level as well. Tables 6 and 7 demonstrate the average content distributions of the CBAS-R as administered to participants in kindergarten, 1st and 2nd grades. In Table 6, all 40 items are considered; in Table 7, the post hoc termination criterion was adjusted to 20 items.

Table 6: *Number of Items from Each Domain, by Grade (termination = 40 items)*

Grade	n	Concepts of Print	Phonemic Awareness	Phonics	Vocabulary	Comprehension
K	76	18.54	6.88	7.72	1.08	0.22
1	95	6.60	2.03	10.76	2.16	2.06
2	116	1.26	0.64	10.50	5.14	6.65
All	287	7.60	2.75	9.85	3.08	3.43

Table 7: *Number of Items from Each Domain, by Grade (termination = 20 items)*

Grade	n	Concepts of Print	Phonemic Awareness	Phonics	Vocabulary	Comprehension
K	76	10.54	3.51	1.74	1.07	0.22
1	95	4.68	1.21	2.31	1.77	1.41
2	116	0.90	0.27	3.04	3.50	3.52
All	287	4.70	1.44	2.45	2.28	1.95

Question 3: Criterion validity

How valid is the CBAS-R ability estimate? Does it appropriately discriminate students across grade levels? Does it correlate to criterion measures, including teacher ratings and reading inventories?

A valid measure discriminates lower performing students from higher on the trait of interest. Because grade level serves as proxie for differences in mean performance on such academic skills as reading, differences in mean performance across grade level for the CBAS-R should follow grade levels. Tables 8 and 9 present final CBAS-R theta estimates by grade level, demonstrating the expected increases across grade level. Table 8 presents the CBAS-R theta estimates with termination set at 40 items, and Table 9 presents them with termination at 20 items.

Table 8. *Final Theta Estimate by Grade (termination = 40 items)*

Grade	n	M	(sd)	Min	Max
K	76	-1.49	(0.57)	-3.00	-0.37
1	95	-0.42	(0.62)	-2.32	1.13
2	116	0.21	(0.62)	-3.00	1.18
All	287	-0.45	(0.91)	-3.00	1.18

Table 9. *Final Theta Estimate by Grade (termination = 20 items)*

Grade	n	M	(sd)	Min	Max
K	76	-1.65	(0.55)	-3.00	-0.24
1	95	-0.55	(0.78)	-2.67	1.02
2	116	0.20	(0.68)	-3.00	1.29
All	287	-0.54	(1.00)	-3.00	1.29

Information from the criterion measures—teacher ratings, reading inventories (Fountas & Pinnel, PALS) and correct words read per minute (wpm)—were correlated with the CBAS-R. Criterion measures available varied by school and grade level.

Tables 10 through 17 show correlations between teacher ratings (fallTchr_Rating) and CBAS-R at 40 items (fall_theta_final) and CBAS-R at 20 items (fall20_theta_final). Teacher ratings are also correlated with other criterion measures available for that class such as words per minute (fall_wpm), Fountas and Pinnel (fall_FnP) and/or PALS (fall_PALS).

Most teacher rating correlations with CBAS-R theta estimates range were moderate to high, positive and significant. Correlation values between teacher rating scores and the 40 item test range from .71 to .87 ($p \leq .001$), with the exception of one classroom (.26, n.s.). Correlation values between teacher rating scores and the 20 item test range from .67 to.816 ($p < .01$), with the exception of one classroom (.29, n.s.).

Table 10. *Teacher Rating and Criterion Measure Correlations for Kindergarten Classroom 1*

		fallTchr_Rating	fall20_theta_finaL	fall_theta_final
fallTchr_Rating	Pearson Correlation	1	.816**	.870**
	N	17	17	17
fall20_theta_finaL	Pearson Correlation	.816**	1	.899**
	N	17	17	17
fall_theta_final	Pearson Correlation	.870**	.899**	1
	N	17	17	17

**. Correlation is significant at the 0.01 level (2-tailed).

Table 11. *Teacher Rating and Criterion Measure Correlations for Kindergarten Classroom 2*

		fallTchr_Rating	fall20_theta_finaL	fall_theta_final
fallTchr_Rating	Pearson Correlation	1	.720**	.804**
	N	15	15	15
fall20_theta_finaL	Pearson Correlation	.720**	1	.941**
	N	15	17	17
fall_theta_final	Pearson Correlation	.804**	.941**	1
	N	15	17	17

**. Correlation is significant at the 0.01 level (2-tailed).

Table 12. *Teacher Rating and Criterion Measure Correlations for 1st Grade Classroom 1*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	Fall_wpm	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.776**	.815**	.811**	.705**
	N	18	18	18	16	16
fall20_theta_finaL	Pearson Correlation	.776**	1	.989**	.678**	.445
	N	18	18	18	16	16
fall_theta_final	Pearson Correlation	.815**	.989**	1	.726**	.480
	N	18	18	18	16	16
fall_wpm	Pearson Correlation	.811**	.678**	.726**	1	.700**
	N	16	16	16	16	16
fall_FnP	Pearson Correlation	.705**	.445	.480	.700**	1
	N	16	16	16	16	16

**. Correlation is significant at the 0.01 level (2-tailed).

Table 13. *Teacher Rating and Criterion Measure Correlations for 1st Grade Classroom 2*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	Fall_wpm	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.770**	.828**	.472*	.937**
	N	19	19	19	19	19
fall20_theta_finaL	Pearson Correlation	.770**	1	.953**	.585**	.756**
	N	19	19	19	19	19
fall_theta_final	Pearson Correlation	.828**	.953**	1	.637**	.860**
	N	19	19	19	19	19
fall_wpm	Pearson Correlation	.472*	.585**	.637**	1	.597**
	N	19	19	19	19	19
fall_FnP	Pearson Correlation	.937**	.756**	.860**	.597**	1
	N	19	19	19	19	19

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Table 14. *Teacher Rating and Criterion Measure Correlations for 1st Grade Classroom 3*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	Fall_wpm	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.674**	.723**	.521*	.904**
	N	18	18	18	17	17
fall20_theta_finaL	Pearson Correlation	.674**	1	.967**	.577*	.614**
	N	18	18	18	17	17
fall_theta_final	Pearson Correlation	.723**	.967**	1	.664**	.703**
	N	18	18	18	17	17
fall_wpm	Pearson Correlation	.521*	.577*	.664**	1	.518*
	N	17	17	17	17	17
fall_FnP	Pearson Correlation	.904**	.614**	.703**	.518*	1
	N	17	17	17	17	17

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Table 15. *Teacher Rating and Criterion Measure Correlations for 2nd Grade Classroom 1*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	Fall_wpm	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.782**	.807**	.529**	.783**
	N	25	25	25	25	25
fall20_theta_finaL	Pearson Correlation	.782**	1	.945**	.625**	.835**
	N	25	25	25	25	25
fall_theta_final	Pearson Correlation	.807**	.945**	1	.579**	.860**
	N	25	25	25	25	25
fall_wpm	Pearson Correlation	.529**	.625**	.579**	1	.716**
	N	25	25	25	25	25
fall_FnP	Pearson Correlation	.783**	.835**	.860**	.716**	1
	N	25	25	25	25	25

**. Correlation is significant at the 0.01 level (2-tailed).

Table 16. *Teacher Rating and Criterion Measure Correlations for 2nd Grade Classroom 2*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.677**	.748**	.891**
	N	24	24	24	24
fall20_theta_finaL	Pearson Correlation	.677**	1	.970**	.846**
	N	24	24	24	24
fall_theta_final	Pearson Correlation	.748**	.970**	1	.882**
	N	24	24	24	24
fall_FnP	Pearson Correlation	.891**	.846**	.882**	1
	N	24	24	24	24

**. Correlation is significant at the 0.01 level (2-tailed).

Table 17. *Teacher Rating and Criterion Measure Correlations for 2nd Grade Classroom 3*

		fallTchr_ Rating	fall20_theta_ finaL	fall_theta_ final	Fall_wpm	fall_FnP
fallTchr_Rating	Pearson Correlation	1	.286	.256	.363	.661**
	N	25	25	25	25	25
fall20_theta_finaL	Pearson Correlation	.286	1	.984**	.150	.233
	N	25	26	26	26	26
fall_theta_final	Pearson Correlation	.256	.984**	1	.189	.225
	N	25	26	26	26	26
fall_wpm	Pearson Correlation	.363	.150	.189	1	.417*
	N	25	26	26	26	26
fall_FnP	Pearson Correlation	.661**	.233	.225	.417*	1
	N	25	26	26	26	26

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Tables 18 – 21 show criterion measure correlations between grades and across grades. Kindergarten classrooms show a statistically significant positive relationship between PALS and the 40 item test ($r = .70$, $p<.001$). Similarly, there is a statistically significant positive relationship between PALS and the 20 item test ($r = .70$, $p<.001$).

First grade classrooms show statistically significant positive relationships between the 40 item test and WPM ($r = .64$, $p<.001$), FnP ($r = .72$, $p<.001$), and PALS ($r = .68$, $p<.001$). This trend is similar for the 20 item test: WPM ($r = .58$, $p<.001$), FnP ($r = .64$, $p<.001$), and PALS ($r = .61$, $p<.001$). For the first grade classrooms in this study, the strongest relationship appears to occur between FnP data and CBAS-R theta estimates.

Second grade classrooms show statistically significant positive relationships between the 40 items test and WPM (r = .28, p<.05), FnP (r = .47, p<.001), and PALS (r = .71, p<.001). However, for the 20 item test, the correlation with WPM is not statistically significant (r = .271, p = 0.55). For second grade classrooms in this study, PALS data had the strongest relationship with CBAS-R theta estimates.

Across all grades, correlations with CBAS-R are highest with FnP data for both 40 and 20 items (r = 0.64, p <.001 for both). In general, CBAS-R shows statistically significant positive relationships with all criterion measures utilized in this study and the criterion measures were also positively correlated. Additionally, the 20 item and 40 item assessments were strongly correlated (r = 0.98, p < .001) across all grades.

Table 18. *Criterion Measure Correlations for Kindergarten Classrooms.*

		fall20_theta_finaL	fall_theta_final	fall_PALS
fall20_theta_finaL	Pearson Correlation	1	.938**	.699**
	N	76	76	41
fall_theta_final	Pearson Correlation	.938**	1	.807**
	N	76	76	41
fall_PALS	Pearson Correlation	.699**	.807**	1
	N	41	41	41

**. Correlation is significant at the 0.01 level (2-tailed).

Table 19. *Criterion Measure Correlations for First Grade Classrooms.*

		fall20_theta_finaL	fall_thet	fall_wpm	fall_FnP	fall_PALS
fall20_theta_finaL	Pearson Correlation	1.000	.969**	.577**	.635**	.608**
	N	95.000	95	52	52	38
fall_theta_final	Pearson Correlation	.969**	1.000	.635**	.720**	.679**
	N	95	95.000	52	52	38
fall_wpm	Pearson Correlation	.577**	.635**	1.000	.588**	.[a]
	N	52	52	52.000	52	0
fall_FnP	Pearson Correlation	.635**	.720**	.588**	1.000	.[a]
	N	52	52	52	52.000	0
fall_PALS	Pearson Correlation	.608**	.679**	.[a]	.[a]	1.000
	N	38	38	0	0	38.000

**. Correlation is significant at the 0.01 level (2tailed).

a. Cannot be computed because at least one of the variables is constant.

Table 20. *Criterion Measure Correlations for Second Grade Classrooms.*

		fall20_theta_finaL	fall_theta	fall_wpm	fall_FnP	fall_PALS
fall20_theta_finaL	Pearson Correlation	1.000	.968**	.271	.475**	.673**
	N	116.000	116	51	75	40
fall_theta_final	Pearson Correlation	.968**	1.000	.277*	.471**	.712**
	N	116	116.000	51	75	40
fall_wpm	Pearson Correlation	.271	.277*	1.000	.558**	.[a]
	N	51	51	51.000	51	0
fall_FnP	Pearson Correlation	.475**	.471**	.558**	1.000	.[a]
	N	75	75	51	75.000	0
fall_PALS	Pearson Correlation	.673**	.712**	.[a]	.[a]	1.000
	N	40	40	0	0	40.000

**. Correlation is significant at the 0.01 level (2tailed).

*. Correlation is significant at the 0.05 level (2tailed).

Table 21. *Criterion Measure Correlations for All Grades.*

		fall20_theta_finaL	fall_theta	fall_wpm	fall_FnP	fall_PALS
fall20_theta_finaL	Pearson Correlation	1.000	.980**	.506**	.635**	.241**
	N	287.000	287	103	127	119
fall_theta_final	Pearson Correlation	.980**	1.000	.508**	.644**	.282**
	N	287	287.000	103	127	119
fall_wpm	Pearson Correlation	.506**	.508**	1.000	.682**	.[a]
	N	103	103	103.000	103	0
fall_FnP	Pearson Correlation	.635**	.644**	.682**	1.000	.[a]
	N	127	127	103	127.000	0
fall_PALS	Pearson Correlation	.241**	.282**	.[a]	.[a]	1.000
	N	119	119	0	0	119.000

**. Correlation is significant at the 0.01 level (2tailed).

a. Cannot be computed because at least one of the variables is constant.

Question 4: Response function

To what extent is the response environment of the test (percent correct) equalized across the range of ability estimates?

An important additional advantage of adaptive testing described in the research literature is that students across the range of the trait measured experience a similar testing environment with regard to response patterns. On a conventional, fixed item test, examinees at the extremes of the trait measured exhibit a pattern of either mostly correct or mostly incorrect answers. An adaptive test should produce a more uniform response function, where all examinees across the range of the trait measured answer correctly approximately .50 plus *c*, where *c* is the guessing parameter.

Table 22 compares the CBAS-R at terminations of 20 and 40 items with regard to percent answered correctly across five bins of final theta estimate. Figures 9 and 10 below show the response function for each termination criterion graphically. At the lowest final theta estimates, examinees experienced a lower rate of correct answers than did higher performing examinees.

Table 22: *Response Patterns (Percent Correct) at Contrasting Terminations, by Final Theta Estimate.*

	20 Items Percent Correct			*40 Items* Percent Correct		
Final theta	n	M	(sd)	n	M	(sd)
-3.0 to -2.0	21	.32	(.11)	19	.28	(.08)
-1.99 to -1.0	87	.63	(.09)	65	.64	(.11)
-0.99 to 0	75	.52	(.11)	93	.62	(.13)
.01 to 1.0	95	.54	(.07)	103	.58	(.05)
1.01 to 2.0	9	.61	(.08)	7	.61	(.03)

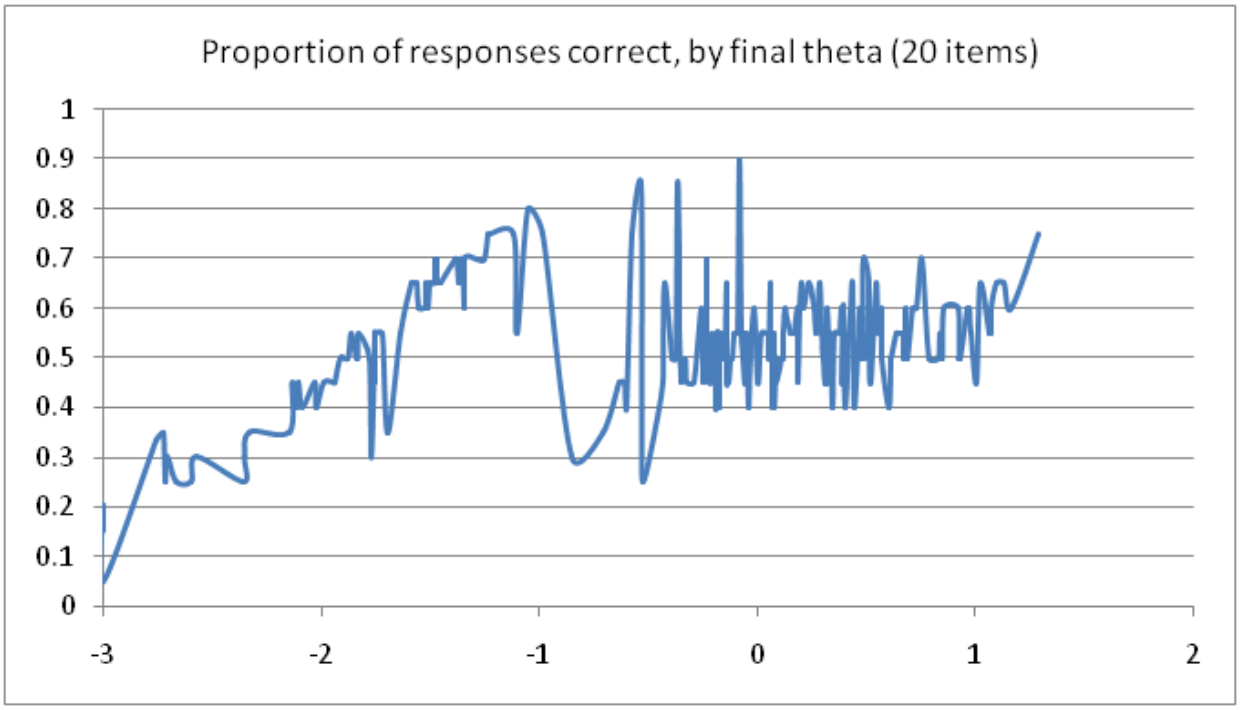


Figure 9. Proportion of Responses Correct, by Final Theta Estimate After 20 items.

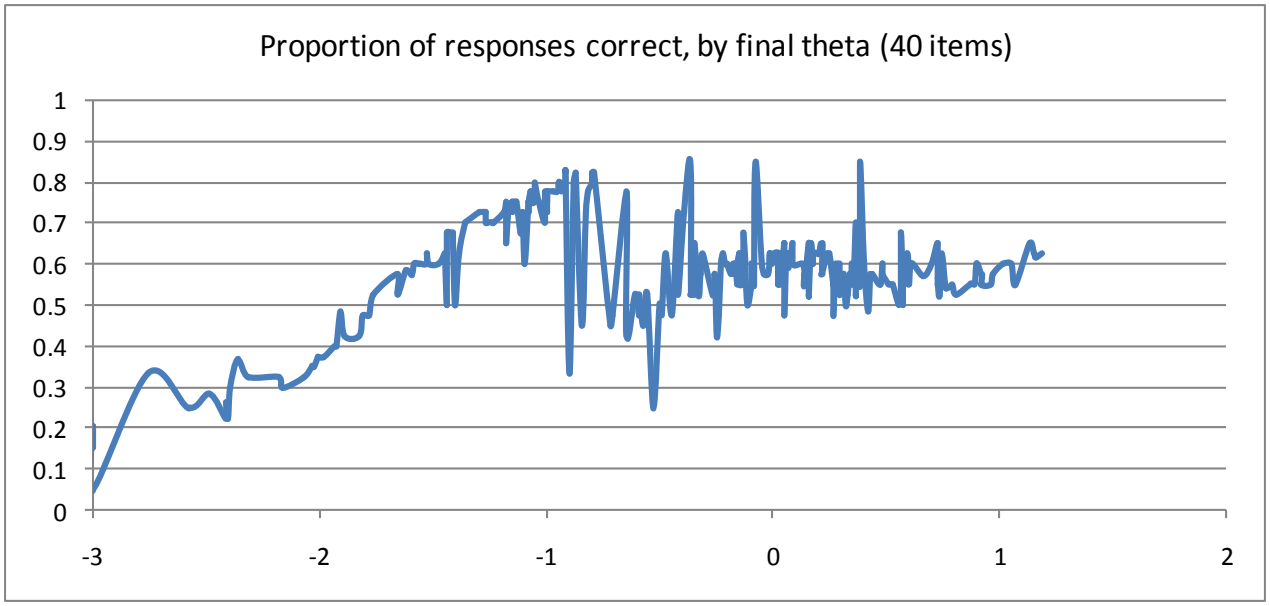


Figure 10. Proportion of Responses Correct, by Final Theta Estimate After 40 items.

The CBAS-R is intended to function across grade levels without producing systematically differential response environments. As demonstrated in Table 23, the average percentage of correct responses did not differ substantially across grade levels.

Table 23. *Response Patterns (Percent Correct) at Contrasting Terminations, by Grade.*

	20 Items			40 Items		
Percent Correct	Percent Correct					
Final theta	n	M	(sd)	n	M	(sd)
K	76	.52	(.15)	76	.55	(.18)
1	95	.57	(.11)	95	.63	(.11)
2	116	.55	(.10)	116	.59	(.10)
All	287	.55	(.19)	287	.59	(.13)

Question 5: Timing

How much time do student examinees take on the assessment?

In order for the CBAS-R to present a feasible option for classroom-based use or for regular progress monitoring, the time each test takes must be sufficiently short. Table 24 demonstrates both the brevity of the CBAS-R and the time savings afforded by setting termination at 20 items as compared to 40. At 20 items, most students take between 2.4 and 8.4 minutes to complete the CBAS-R.

Table 24. *Timing of CBAS Screening CAT at Contrasting Terminations, by Grade.*

	20 items = termination				40 items = termination			
	Total Test Minutes			Seconds Per Item	Total Test Minutes			Seconds Per Item
Grade	**M** (SD)	Min	Max	**M** (SD)	**M** (SD)	Min	Max	**M** (SD)
K	**5.79** (3.98)	2.18	19.87	**19.33** (20.93)	**10.57** (4.67)	4.33	20.07	**19.59** (22.09)
1	**4.49** (2.08)	2.03	14.41	**13.48** (6.25)	**8.67** (3.33)	4.52	20.07	**13.05** (5.19)
2	**5.82** (2.75)	1.82	15.25	**17.45** (8.25)	**10.92** (4.26)	4.17	24.55	**16.63** (7.08)
All	**5.37** (3.00)	1.82	19.87	**16.63** (12.67)	**10.08** (4.20)	4.17	24.55	**16.23** (12.78)

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Appendix A

Teacher Rating Scale for Students' Reading Proficiency

For each student below, please rate his or her ***general proficiency in reading*** relative to other students in your class.

Try to spread student ratings across the ***full range of the scale***, not clustering students only in the middle or toward one end.

Thank you for your help!

Student Name	1 *(least proficient)*	2	3	4	5	6	7 *(most proficient)*
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7

	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7

Appendix B

Items Sample from Each Domain

Phonics

ID: 2333

fat say ban

dock saw

Audio: *"Click on the /a/ sound, as in the word /cake/"*

Comprehension

ID: 2167

Grandpa bought a chair that needed to be assembled. The instructions said to locate the correct screws and bolts, connect the seat and back of the chair, hammer the legs into the seat of the chair, and then lay the cushion on the seat.

After connecting the seat and back of the chair, what should Grandpa do next?

1. Locate the correct screws and bolts.

2. Hammer the legs into the seat of the chair.

3. Put the cushion on the back of the chair.

4. Throw away the chair legs.

Audio: *"Read the sentences and choose the best answer."*

Vocabulary

ID: 2220

The principal addressed the kids in the gym.

_____

lectured stopped left

located happy

Audio: *"Read the sentence. Click on the word that means*

the same as the underlined word."

Fluency/Decoding

ID: 2562

a at on

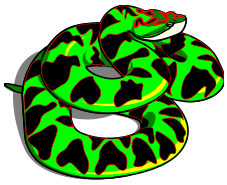
and is

Audio: *"Click on the word 'at'"*

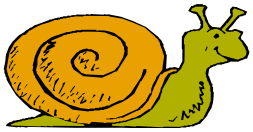
Phonemic Awareness

ID: 1845









Audio: *"Click on the picture that rhymes with cake: key, snake, coat, snail."*